

DotMeet Backend Software Business Requirements Specification (SBR5)

Project Name: DotMeet Backend

Version: 1.0

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1. Introduction

1.1 Purpose

DotMeet is a multi-tenant AI-powered appointment booking and business management SaaS platform. It enables service-based businesses to create their own public booking page, manage appointments, communicate with customers through an AI assistant, and grow their business from a centralized dashboard.

The platform supports multiple business categories including:

- Doctors
- Dentists
- Salons & Barbershops
- Hospitals & Clinics
- Lawyers
- Consultants
- Tutors
- Fitness Trainers
- Carpenters
- Electricians
- Plumbers
- Freelancers
- Agencies
- Any appointment-based service provider

Each business owner receives a unique public page where customers can learn about their services, chat with an AI assistant, and book appointments online.

1.2 Scope

The backend system will provide:

Authentication & Authorization

- Email Registration
- Google Authentication
- OTP Verification
- JWT Authentication
- Refresh Tokens
- Role-Based Access Control (RBAC)

Business Owner Management

- Business Registration
- Business Category Selection
- Username Creation
- Unique Public Booking Page
- Business Verification
- Business Approval by Admin

Example:

<https://dotmeet.com/dr-shahin>

<https://dotmeet.com/smile-dental>

<https://dotmeet.com/elegant-salon>

Public Business Page

Each business page includes:

- Business Profile
- About
- Services
- Pricing Plans
- Gallery
- Working Hours
- Contact Information
- Google Map
- Reviews
- Social Media Links
- Online Booking
- AI Assistant

Appointment Management

- Create Appointment

- Update Appointment
- Cancel Appointment
- Reschedule Appointment
- Manual Appointment by Business Owner
- Online / Offline Appointment
- Booking Calendar
- Booking History

AI Assistant

Every business has its own AI assistant.

The AI can:

- Answer customer questions
- Explain services
- Suggest packages
- Recommend pricing
- Book appointments
- Collect customer information
- Generate leads
- Support Bangla & English conversations

The AI only accesses the current business owner's information using the username from the public URL.

Service Management

Business owners can:

- Create Services
- Set Pricing
- Service Duration
- Service Images
- Availability
- Categories
- Service Status

Subscription System

Admin manages subscription plans.

Plans may include:

- Free
- Starter

- Professional

Each plan controls:

- Number of Services
- AI Access
- Gallery Limit
- Booking Limit
- Storage
- Analytics
- Custom Domain

Payment Gateway

Support:

- SSLCommerz
- AamarPay
- Stripe (Future)

Admin Management

Admin can manage:

- Users
- Business Owners
- Categories
- Bookings
- Payments
- Transactions
- AI Conversations
- Support Tickets
- Subscription Plans
- Coupons
- Reports
- Website Settings
- About Us, FAQs, Terms & Conditions, Privacy Policy, blogs, banners, etc.

1.3 Definitions

JWT – JSON Web Token

RBAC – Role-Based Access Control

CRUD – Create, Read, Update, Delete

OTP – One-Time Password

REST API – Representational State Transfer API

AI – Artificial Intelligence

SaaS – Software as a Service

2. Overall Description

2.1 Product Perspective

DotMeet is a cloud-based multi-tenant SaaS platform.

A single backend serves thousands of businesses while securely isolating each business's data.

The backend exposes REST APIs consumed by Next.js, React

2.2 User Roles

Visitor

- Browse business profiles
- Search businesses
- Chat with AI
- View services
- Book appointments

Customer

- Register/Login
- Book appointments
- Pay online for buy services
- Review services
- View booking history

Business Owner

- Manage business profile
- Manage services
- Manage bookings
- Manage working hours
- Upload gallery

- Generate invoices
- Configure AI assistant
- View analytics
- Create support tickets for admin
- Create manual appointments

Admin

- Manage users
 - Approve businesses
 - Manage subscriptions
 - Manage payments
 - Manage categories
 - View analytics
 - Resolve support tickets
 - Manage AI settings
 - Configure payment gateways
 - Monitor platform activity
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3. Functional Requirements

Authentication

FR1: User registration using Email, Google, and Facebook

FR2: OTP verification before account activation.

FR3: Login using Email, Google, and Facebook

FR4: Forgot Password.

FR5: Change Password.

FR6: Refresh Access Token.

User Management

FR7: Update Profile.

FR8: Upload Profile Image.

FR9: Admin manages users.

FR10: Activate or deactivate accounts.

Business Owner

FR11: Manual Register business.

FR12: Select business category.

FR13: Reserve unique username.

FR14: Generate public business page.

FR15: Submit business for approval.

FR16: Admin approves or rejects business.

Service Management

FR17: Create service.

FR18: Update service.

FR19: Delete service.

FR20: Configure pricing.

FR21: Upload service images.

FR22: Configure service duration.

Appointment Management

FR23: Search businesses.

FR24: View available services.

FR25: Book appointment.

FR26: Select preferred date and time.

FR27: Reschedule booking.

FR28: Cancel booking.

FR29: Manual booking by business owner.

FR30: Calendar management.

FR31: Booking reminders.

Payment

- FR32: Pay using SSLCommerz.
 - FR33: Pay using AamarPay.
 - FR34: Payment verification.
 - FR35: Payment success/failure callbacks.
 - FR36: Invoice generation.
 - FR37: Refund management (Future).
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AI Assistant

- FR38: Business-specific AI chatbot.
 - FR39: AI answers FAQs.
 - FR40: AI recommends services.
 - FR41: AI books appointments.
 - FR42: AI collects customer leads.
 - FR43: AI stores conversation history.
 - FR44: AI prompt customization by business owner.
 - FR45: AI supports Bangla and English.
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Reviews

- FR46: Customers can review completed appointments.
 - FR47: Business owners can respond to reviews.
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Support

- FR48: Customer support tickets.
- FR49: Business owner supports tickets.
- FR50: Admin support management.

Subscription

FR51: Subscribe to plans.

FR52: Renew subscriptions.

FR53: View remaining subscription days.

FR54: Upgrade or downgrade plans.

Analytics

FR55: Dashboard statistics.

FR56: Revenue reports.

FR57: Booking analytics.

FR58: AI conversation analytics.

FR59: Customer insights.

4. Business Owner Dashboard

Modules include:

- Dashboard Overview
- Business Profile
- Services
- Pricing
- Gallery
- Working Hours
- Calendar
- Appointment Management
- Manual Booking
- Customer Management
- Reviews
- AI Assistant Settings
- AI Prompt Configuration
- Conversation History
- Lead Management

- Staff Management
 - Analytics
 - Subscription
 - Invoice Management
 - SEO Settings
 - Contact Information
 - Support Tickets
 - Business Settings
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5. Admin Dashboard

Modules include:

- Dashboard
 - Users
 - Business Owners
 - Business Categories
 - Business Approval
 - Services
 - Bookings
 - Payments
 - Transactions
 - Subscriptions
 - Subscription Plans
 - Coupons
 - AI Conversations
 - AI Usage Analytics
 - Support Tickets
 - Reviews
 - CMS
 - Email Templates
 - Notification Templates
 - Payment Gateway Settings
 - Website Settings
 - Reports
 - Audit Logs
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6. Non-Functional Requirements

NFR1: 95% of API requests should respond within 500ms.

NFR2: Support at least 1,000 concurrent users.

NFR3: Horizontal scalability.

NFR4: Password hashing using bcrypt.

NFR5: Secure JWT authentication.

NFR6: RBAC enforcement.

NFR7: HTTPS for all communication.

NFR8: 99.5% system uptime.

NFR9: Redis for OTP, caching, and session management.

NFR10: Modular architecture following Clean Architecture principles.

NFR11: API versioning (/api/v1).

NFR12: Comprehensive logging and monitoring.

7. System Architecture

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose
- Redis
- JWT

Frontend

- Next.js
- Redux Toolkit

Cloud Storage

- Cloudinary

Payment

- SSLCommerz
- AamarPay

AI

- OpenAI / OpenRouter

Deployment

- VPS
 - AWS
 - DigitalOcean
 - Docker
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8. Data Requirements

Core Entities

- Users
 - Businesses
 - Business Categories
 - Services
 - Service Orders
 - Appointments
 - Working Hours
 - Staff
 - Galleries
 - Reviews
 - Payments
 - Transactions
 - Subscriptions
 - AI Conversations
 - AI Leads
 - Notifications
 - Support Tickets
 - Analytics
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9. Example Use Case

Booking an Appointment

1. Customer opens a business page.
 2. AI assistant answers customer questions.
 - 3.
 4. Customer selects a service.
 5. Customer chooses a date and time.
 6. Booking is created with **Pending** status.
 7. Business owner receives a notification.
 8. Customer receives email/SMS confirmation.
 9. Business owner changes Appointment status to **Confirmed to complete**.
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10. Acceptance Criteria

The system will be considered complete when:

- Users can register and authenticate securely.
- Business owners can create and manage public business pages.
- Visitors can search businesses and book appointments.
- AI assistant answers business-specific questions and supports booking.
- Online payments are processed successfully.
- Business owners can manage services, bookings, and customers.
- Admin can manage the entire platform.
- APIs are secure and role-based.
- The platform meets defined performance, scalability, and security requirements.